Do Now · retrieval

Lesson 132 · Hybrid Encryption: Protecting a Connection

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which key verifies a signature produced using an owner's private key?

Q2

Which memory is typically volatile: RAM or ROM?

Q3

In a queue, what does FIFO stand for?

Q4

What is O(n) search complexity associated with?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Authenticate and agree — Client validates server identity and securely establishes shared key material.

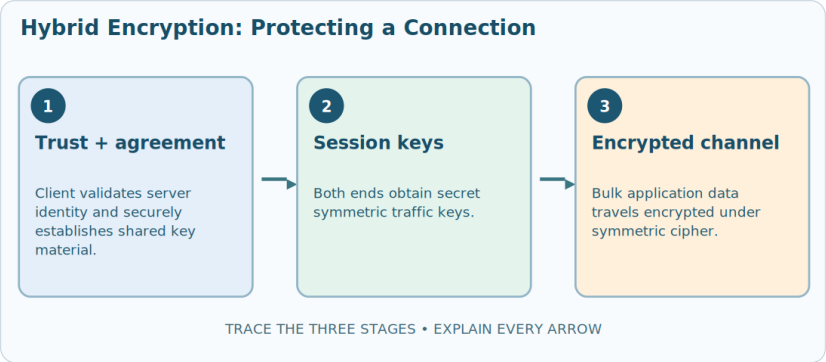
Core idea 2

Derive session key — Both ends obtain secret symmetric traffic keys.

Core idea 3

Encrypt traffic — Bulk application data travels encrypted under symmetric cipher.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

session key — Temporary symmetric key for a connection/session.

hybrid encryption — Combines asymmetric key establishment with symmetric data encryption.

certificate — Credential binding identity to a public key, subject to trust validation.

confidentiality — Preventing unauthorised reading of information.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

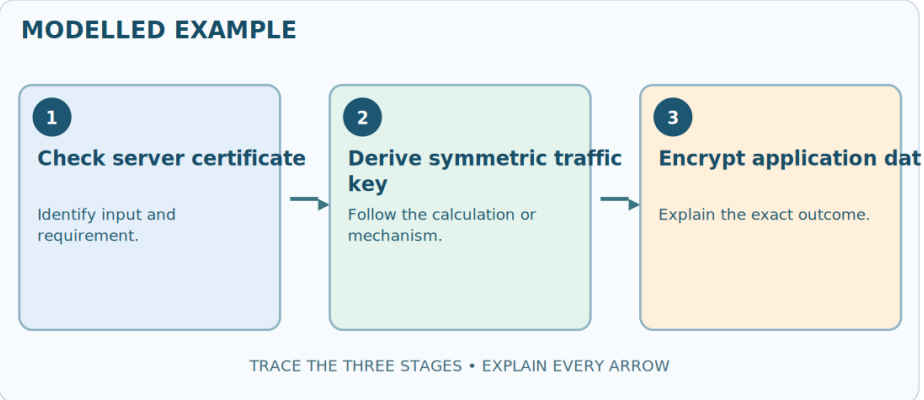
Follow the worked method

1. Check server certificate — Predict the first step from the prompt.

2. Derive symmetric traffic key — Explain the intermediate result or decision.

3. Encrypt application data — Modern secure connections typically authenticate with public-key credentials, establish shared session keys (often via ephemeral key agreement) and use efficient symmetric encryption for bulk traffic. Identity checks guard against impostors.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

HTTPS is encrypted
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe why hybrid cryptography uses asymmetric and symmetric methods together. [6]

Question 2

Explain what a session key is and why it is not the same as a website's public key. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate the assertion 'the padlock means any entered data are safe'. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

HTTPS guarantees the website operator is honest.

Use key vocabulary from memory

1. Explain session key in your own words.

2. Explain hybrid encryption in your own words.

3. Explain certificate in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.